



Indywidualny Plan Badawczy (IPB) Individual Research Plan (IRP)

1. Podstawowe dane / Basic data

Podstawowe informacje dotyczące doktoranta i realizowanej rozprawy doktorskiej. W przypadku promotora spoza Politechniki Wrocławskiej należy podać afiliację. / *Basic information on the doctoral student and the doctoral dissertation underway. In the case of a supervisor from outside Wrocław University of Science and Technology, affiliation should be provided.*

Imię i nazwisko / Name, surname	Grzegorz Piotrowski
Dyscyplina kształcenia / Discipline of education	informatyka techniczna i telekomunikacja / information and communication technology
Wydział / Faculty	W4 Wydział Informatyki i Telekomunikacji / Faculty of Information and Communication Technology
Katedra / Department	K46 Katedra Sztucznej Inteligencji / Department of Artificial Intelligence
Data rozpoczęcia kształcenia / Date of commencement of the doctoral education	01.10.2025
Nr ORCID	0009-0004-6013-0461
Promotor/ promotorzy / Supervisor(s)	dr hab. inż. Tomasz Jan Kajdanowicz, prof. uczelni, Politechnika Wrocławska (K46, W4), ORCID 0000-0002-8417-1012
Promotor pomocniczy / Assistant supervisor	—

2. Temat rozprawy doktorskiej / Topic of doctoral dissertation

Propozycja tematu rozprawy doktorskiej. Przedstawiony temat nie musi stanowić ostatecznego tytułu, jaki zostanie nadany rozprawie doktorskiej złożonej na zakończenie kształcenia w Szkole Doktorskiej. / *Proposed topic for a doctoral dissertation. The presented topic does not have to constitute the final title of the doctoral dissertation submitted at the end of education at the Doctoral School.*

PL: Uczenie polityk agentów ucieleśnionych w pętli rzeczywistość–symulacja–rzeczywistość przy ograniczonym budżecie danych rzeczywistych z wykorzystaniem cyfrowych bliźniaków z neuronowej rekonstrukcji scen

EN: Learning embodied policies in the real-to-sim-to-real loop under a limited real-data budget with digital twins from neural scene reconstruction

3. Harmonogram przygotowania rozprawy doktorskiej / Schedule for the preparation of the doctoral dissertation

Harmonogram przygotowania rozprawy doktorskiej z wyszczególnieniem etapów i miejsc prowadzonych badań, w tym terminów zakończenia badań cząstkowych (semestr kształcenia) i opracowania ich wyników oraz opcjonalnie w zależności od rodzaju prowadzonych badań, wybranych między innymi spośród następujących:



- realizacja badań (w tym przygotowanie stanowiska badawczego, przygotowanie materiałów do badań, wykonanie pomiarów/eksperymentów, opracowanie otrzymanych wyników),
- przygotowanie i wysłanie artykułu naukowego do redakcji czasopisma naukowego,
- wysłanie zgłoszenia patentowego do właściwego urzędu,
- opracowanie propozycji wdrożenia dla przemysłu,
- udział w konferencjach, warsztatach lub szkołach letnich/zimowych i prezentacja wyników badań,
- przygotowanie wniosku i wystąpienie o grant,
- mobilność krajowa lub zagraniczna doktoranta (w tym konsultacje naukowe i wizyty studyjne w ośrodkach krajowych lub zagranicznych).

A schedule for the preparation of the doctoral dissertation detailing the stages and locations of the research to be conducted, including the dates for completing the sub-studies and developing their results, and optionally, depending on the type of research to be conducted, selected among the following:

- *execution of the research (including preparation of the research site, preparation of materials for the research, execution of measurements/experiments, elaboration of the obtained results),*
- *preparation and submission of a scientific article to the editor of a scientific journal,*
- *submission of a patent application to the relevant office,*
- *development of an implementation proposal for the industry,*
- *participation in conferences, workshops or summer/winter schools and presentation of research results,*
- *preparation of an application and application for a scholarship,*
- *domestic or foreign mobility of the doctoral student (including scientific consultations and study visits to domestic or foreign centres).*

Okres realizacji zadania (semestr kształcenia) / Task implementation period (semester)	Zwięzły opis zadania / Brief description of the task
Semestr 1	<i>(Oct 2025 – Feb 2026)</i> Research target: define the problem and map the state of the art. T1.1 Systematic literature review in four areas: (a) sim-to-real transfer and domain randomization, (b) learning under distribution shift and domain adaptation, (c) neural scene reconstruction (NeRF, 3D Gaussian Splatting), (d) learning-based visual navigation. <i>D:</i> annotated bibliography and a written research-gap statement (basis of §6). T1.2 Identification of the open problem: how much real data sim-to-real transfer needs, and whether reconstructed scenes can replace real data. <i>D:</i> problem statement and preliminary research questions, discussed with the supervisor. T1.3 Doctoral School courses of semester 1. <i>D:</i> completed courses. Place: K46.



Semestr 2	(Mar – Sep 2026) Research target: formulate the research plan and select the experimental basis. T2.1 Formulation of the data-efficient real-to-sim-to-real thesis, research questions RQ1–RQ4 (capture less, train robustly on an imperfect twin, collect few real rollouts, budget of the whole loop) and hypotheses H1–H4 with measurable decision rules and evaluation tiers A–C. <i>D:</i> §7 and §9 of this IPB. T2.2 Comparative survey of public benchmarks and datasets (e.g. ScanNet++, HM3D), simulators (e.g. Habitat-Sim) and 3D Gaussian Splatting implementations against the needs of RQ1–RQ4 (real video + reference scan, licence, compute cost). <i>D:</i> tool-selection note with a shortlist for Stage I. T2.3 Individual Research Plan agreed with the supervisor. <i>D:</i> this IPB, signed and submitted by 30.09.2026. Place: K46.
Semestr 3	(Oct 2026 – Feb 2027) Stage I – capture less (RQ1, H1). T3.1 Tier A set-up: final choice of ≥ 10 public scenes with dense real captures and reference scans, simulator and existing 3DGS pipeline from the T2.2 shortlist; reference twins ("reality") and training twins at ≥ 4 capture budgets; navigation policy training in the twins; pre-registration of the protocol (§7, §9). <i>D:</i> working loop and protocol in the repository, by Dec 2026. Place: K46, WCSS. T3.2 Validation stand: written agreement on robot access (planned cooperation with the K29 Denali laboratory, or another PWR platform) and sensor decision (SzD Minigrant application if a sensor must be bought). <i>D:</i> agreement, by Nov 2026. T3.3 Task-aware, uncertainty-guided capture vs. uniform and reconstruction-only view selection. <i>D:</i> first budget–success curves (first H1 result), by Feb 2027.
Semestr 4	(Mar – Sep 2027) Stage I completed; Stage II – train robustly on an imperfect twin (RQ1–RQ2, H1–H2). T4.1 Uncertainty-aware augmentation and representation-based weighting of twin data vs. uniform domain randomization at an equal capture budget. <i>D:</i> H1 completed and first H2 result on tier A, by Apr 2027; H2 completed, by Jul 2027. Place: K46, WCSS. T4.2 Prepare and submit article P1 to NeurIPS 2027 (task-aware capture and uncertainty-aware training, RQ1–RQ2). <i>D:</i> submission, May 2027 (§11). T4.3 Stage III set-up: rollout-selection and correction code on tier A (Jul–Sep 2027). <i>D:</i> first H3 result, by Sep 2027. T4.4 Grants and international activity: NCN PRELUDIUM application; poster at an international summer school; foreign co-author sought for P2; NAWA Bekker application for the sem. 6 visit. <i>D:</i> applications, poster.



Semestr 5	(Oct 2027 – Feb 2028) Mid-term evaluation (Nov 2027): autoreferat (≤ 5 pages, ~mid-Oct 2027) and 15-min presentation. Stage III – collect few real rollouts (RQ3, H3). T5.1 Active selection of real rollouts (predicted gap, uncertainty) vs. random selection; correction of the twin and fine-tuning of the policy with the selected rollouts. <i>D:</i> H3 completed on tier A, by Feb 2028. Place: K46, WCSS. T5.2 First robot validation campaign (tier C, 2 PWr rooms, ~8 robot-hours): own captures, twins, uniform capture vs. the loop. <i>D:</i> validation report. Place: Denali. Without robot access, the campaign moves to sem. 7; H1–H3 are decided on tier A either way. T5.3 Article P2 (active real-rollout selection and twin correction, RQ3) submitted to CVPR 2028 (~Nov 2027; ICLR 2028, ~late Sep 2027, if ready earlier). If P1 was rejected, its revision goes to CVPR 2028 or ICLR 2028. <i>D:</i> submission(s).
Semestr 6	(Mar – Sep 2028) Stage IV – budget of the whole loop; cross-task test (RQ4, H4). T6.1 (a) Real-data budget curves of the full loop vs. real-only learning ("exchange rate") on tier A; (b) the same pipeline, unchanged, on manipulation scenes, with published paired sim/real results as tier-B check. <i>D:</i> navigation result by Apr 2028; H4 completed, by Sep 2028. Place: K46, WCSS. T6.2 Foreign research visit (3 months, NAWA Bekker; if not funded, a 5–30-day Erasmus+ short-term mobility to a European lab; hosts in §12): joint experiments. <i>D:</i> visit report. T6.3 Article P3 (budget law of the full loop, with first manipulation results) submitted to NeurIPS 2028 (~May 2028); if not ready, to ICLR 2029 (~late Sep 2028). <i>D:</i> submission.
Semestr 7	(Oct 2028 – Feb 2029) Consolidation (RQ1–RQ4). T7.1 Second robot validation campaign (tier C) of the full loop; final analysis of H1–H4 across tiers. <i>D:</i> final results. Place: K46, Denali. T7.2 Release of code and experiment scripts. <i>D:</i> public repository. T7.3 Optional article P4 (consolidated study) to ICLR 2029 (~late Sep 2028) or CVPR 2029 (~Nov 2028); start of dissertation writing. <i>D:</i> submission (optional), dissertation outline.
Semestr 8	(Mar – Sep 2029) T8.1 Editing of the doctoral dissertation; final version. <i>D:</i> dissertation submitted by 30.09.2029 (§4). T8.2 Responses to reviews and camera-ready versions of pending articles. <i>D:</i> revised manuscripts.

4. Termin złożenia rozprawy doktorskiej / Date of the doctoral dissertation submission

Planowany termin złożenia rozprawy doktorskiej nie może przekraczać daty zakończenia ósmego semestru kształcenia w Szkole Doktorskiej. / *A planned date for submission of the dissertation, which must not exceed the date of completion of the eighth semester of education in the Doctoral School.*

30.09.2029



(rekomendowana data to ostatni dzień z okresu 4 lata kształcenia w Szkole Doktorskiej, np. jeżeli kształcenie rozpoczęło się 1.10.2025, to proszę wpisać 30.09.2029 /

The recommended date, which marks the final day of the four-year doctoral training period. For example, if your training began on October 1st, 2025, please enter September 30th, 2029)

5. Uzasadnienie wyboru tematu rozprawy doktorskiej / Justification for choosing the topic of the doctoral dissertation

Opis zawierający uzasadnienie wyboru tematu rozprawy doktorskiej uwzględniający potencjalne obszary zastosowania planowanych wyników rozprawy doktorskiej (maks. 1 strona, czcionka 11 pt., interlinia 1). / Description containing the justification for the selection of the topic of the doctoral dissertation, taking into account the potential areas of application of the planned results of the doctoral dissertation (at most 1 page, font 11, line spacing 1).

Modern machine learning is limited less by models than by **data in the target domain**. Embodied agents (mobile robots, manipulators) learn strong behaviour only from very large amounts of experience. In the real world such data is slow, expensive and sometimes risky to collect, so a policy is usually trained in simulation and then deployed on real sensor data, where it often fails (the **sim-to-real gap**).

Digital twins built by neural scene reconstruction (3D Gaussian Splatting, Neural Radiance Fields) turn a short real capture into a photorealistic model of the target scene, in which a policy can be trained at scale. This gives a **real-to-sim-to-real loop**: capture the real scene, build a twin, train the policy in the twin, run a few real trials, correct the twin and the policy, deploy. Building twins has quickly become routine, with dozens of papers a year and more than ten open-source twin simulators. What remains open is the **real data** the loop consumes. Each step spends it: the capture, the real data used to make training robust to what the twin got wrong, and the real trials used to check and correct the policy. Today these amounts are chosen by habit: the scene is recorded uniformly, the policy is trained with generic randomization, and real trials are picked at random or by hand. How much real data a twin actually saves, especially for navigation, has not been measured.

What the dissertation proposes. The real data needed by the loop can be reduced by **allocating it actively**: capture only what the policy needs to build the twin, train so that the policy is robust to the twin's errors, and collect only the few real rollouts that close the remaining gap. Two signals, both available without extra real data, guide these choices: the **uncertainty of the reconstruction** (where the twin is unreliable) and the **policy's own internal representations** (where twin and real inputs look different to the policy). The result is measured as a budget–performance curve: how much real data the loop needs to reach a given real success rate, compared with uniform allocation and with learning from real data only.

Why this topic, and why in this discipline. The object is a data-efficient learning methodology (active learning, learning under distribution shift, uncertainty estimation), not a robot, a simulator or a benchmark, which places it in *information and communication technology*. The student uses existing twin pipelines, simulators and public datasets and does not develop new ones. **Visual navigation** is the primary testbed, because public scene datasets with dense real captures make it possible to count real data exactly without an own robot; **robotic manipulation** tests that the methods carry over to another task. Trials on a real robot (planned cooperation with a PWr robotics laboratory) validate, but do not decide, the conclusions. The topic fits the Department of Artificial Intelligence (K46), whose research groups include representation learning. The computation runs on the GPU infrastructure available to PWr researchers (WCSS, PLGrid).

Potential application areas: faster and cheaper deployment of learned robots in new buildings and workcells (service, assistive, warehouse and inspection robotics, flexible manipulation), with less time spent recording scenes and supervising real trials; guidance on how to record a site for a twin; and, beyond robotics, any system trained on reconstructed or synthetic data and deployed on real sensor



data (industrial digital twins, perception for autonomous systems), where real data must be collected on a budget.

6. Zarys aktualnego stanu badań w tematyce rozprawy doktorskiej / Outline of the current state of research in the topic of the doctoral dissertation

Przegląd literaturowy zawierający opis aktualnego stanu badań w tematyce rozprawy doktorskiej uwzględniający odwołania do najważniejszych pozycji literaturowych związanych z tematyką rozprawy doktorskiej (maks. 2 strony, czcionka 11 pt., interlinia 1). / A literature review containing a description of the current state of research in the subject of the doctoral dissertation, including references to the most important literature items related to the subject of the doctoral dissertation (at most 2 pages, font 11, line spacing 1).

Real-to-sim-to-real with neural digital twins. Simulators such as Habitat [1] made large-scale training of embodied agents possible, but a generic simulator is not the robot's scene. 3D Gaussian Splatting (3DGS) [2] now turns a short camera capture into a real-time, photorealistic *digital twin* of that scene. Policies trained in such twins transfer to real robots in navigation (EmbodiedSplat [3], Vid2Sim [4], GaussGym [5]) and manipulation (RialTo [6], SplatSim [7]), and GPU simulators ship twin environments (ManiSkill3 [8]). Building a twin is no longer the bottleneck; **real data** is: the capture that builds the twin and the real rollouts that correct the policy. Current systems fix this cost by hand. EmbodiedSplat uses a 20–30-minute phone capture per scene, and RialTo builds its twins from "small amounts of real-world data"; its ablation over 0–15 real demonstrations is the closest published analysis of a real-data budget.

How much real data does transfer need? Real-only scaling studies relate performance to the number of real demonstrations in manipulation [9]; in navigation, diversity matters more than quantity and more data from a known location saturates quickly [10]. On the twin side, CASHIER [11] reports performance that scales super-linearly with human effort, X-Sim [12] matches behaviour cloning with ten times less data-collection time, and sim-and-real co-training raises real success by 38% on average [13]. Simulation can poorly predict real navigation results (Sim-vs-Real Correlation Coefficient, SRCC [14]), and lower-fidelity simulation can transfer better [15], so more capture is not automatically better. All of these report results at fixed or hand-chosen budgets. **We found no study that varies the capture budget of a twin as an experimental variable in navigation, or that fits real-data budget curves for the whole loop against real-only learning.**

Capturing less: active reconstruction and its uncertainty. Active view selection chooses the images that improve a reconstruction most: FisherRF [16] maximizes the expected information gain on radiance-field parameters, GenNBV [17] learns a next-best-view policy that generalizes across scenes, and Bayes' Rays [18] estimates a volumetric uncertainty field for a trained neural radiance field. Their objective is reconstruction quality or coverage, not what a downstream policy needs; task-aware capture for twins has only 0–2 Semantic Scholar hits per year (2023–2026). ScanNet++ [19] pairs real captures of rooms with reference laser scans, so capture budgets can be varied offline.

Training robustly on an imperfect twin. Domain randomization varies appearance [20] or dynamics [21] uniformly within hand-set ranges, and SimOpt [22] adapts the randomization distribution from a few real rollouts. Domain adaptation aligns sim and real features in co-training [23], and a mechanistic analysis finds that representation alignment is the main effect of co-training [24]. Frozen encoders such as DINOv2

[25] allow twin and real frames to be compared, and Phys2Real [26] trains with uncertainty over physical

parameters. Still, a twin's own **reconstruction uncertainty** is rarely used as a training signal (0–1 hits per year), and weighting twin samples by their representation distance to a small real set appeared in one paper in 2023–2026.



Collecting few real rollouts. Active learning selects the most informative real samples for sim-to-real adaptation in grasping [27], AMF [28] chooses which tasks to demonstrate under a demonstration budget, and active experiment selection reduces the cost of real evaluation [29]. CUPID [30] and DataMIL [31] select training data by its influence on closed-loop success. Prediction-powered inference [32] combines many cheap predictions with a few labels, SureSim [33] applies it to simulated and real trials, and SIMPLER

[34] shows that simulated evaluation of manipulation policies can track real evaluation. GaussTwin [35]

corrects a 3DGS twin from photometric error. These methods select demonstrations, tasks or trials; none selects the real rollouts that should correct a reconstructed twin and its policy.

Research gap. Real-to-sim-to-real works, but its real-data cost is set by hand and spent uniformly. Missing are (i) task-aware, uncertainty-guided capture judged by downstream policy success rather than image quality (RQ1), (ii) training that uses the twin's reconstruction uncertainty and its representation distance to a small real set (RQ2), (iii) active selection of the few real rollouts that correct the twin and the policy (RQ3), and (iv) a total real-data budget curve of the full loop against real-only learning, in navigation and manipulation (RQ4). This dissertation addresses that gap.

References

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7. Pytania i hipotezy badawcze / Research questions and hypotheses

Opis najważniejszych pytań i hipotez badawczych związanych z realizowaną rozprawą doktorską (maks. 1 strona, czcionka 11 pt., interlinia 1). / Description of the most important research questions and hypotheses related to the doctoral dissertation (at most 1 page, font 11, line spacing 1).

Thesis. The real data needed in a real-to-sim-to-real loop can be reduced substantially by **allocating it actively**: capture only what the policy needs to build the twin, train so that the policy is robust to what the twin got wrong, and collect only the few real rollouts that close the remaining gap. Reconstruction uncertainty and the policy's own representations guide all three steps. Testbeds: navigation (primary), manipulation (cross-task).

Tier A protocol (proxy reality). On ≥ 10 public indoor scenes with dense real captures and reference scans (e.g. ScanNet++, some held out), a **high-fidelity reference built from the full capture plays "reality"**; a 3D Gaussian Splatting *twin* built from a subset of the real views is the training simulator. Real data is counted exactly: capture = real views given to the twin, real rollouts = episodes in the reference, summed in one operator-time cost at a fixed rate. SR = success rate in the reference. Tier A decides; tier B (public real-world data, published paired sim/real results) and tier C (real robot) only report agreement. Tests are one-sided, $\alpha = 0.05$, bootstrapped over scenes and seeds, Holm-corrected within each hypothesis and pre-registered with all thresholds.

RQ1 (real $\rightarrow$ sim: capture less). How little capture does a twin good enough for policy learning need, and can the task guide the capture?

H1. Task-aware, uncertainty-guided capture (next views chosen where task-relevant regions have high reconstruction uncertainty) reaches the SR of uniform capture with $\geq 40\%$ fewer views. *Decision:* on fitted capture-SR curves, $C_{\text{task}}(\tau_1) \leq 0.6 \cdot C_{\text{uniform}}(\tau_1)$, and the upper 95% bound of $C_{\text{task}} / C_{\text{recon}}$ (reconstruction-only view selection) is < 1 ; τ_1 = SR of uniform capture at its largest budget.

RQ2 (in sim: train robustly on an imperfect twin). How should a policy be trained to be robust to the twin's reconstruction errors?

H2. Uncertainty-aware training (augmentation and sample weights from per-region reconstruction uncertainty and from the representation distance to a few real images) improves real transfer over uniform domain randomization at an equal capture budget. *Decision:* mean paired SR gain ≥ 10 pp on held-out scenes, lower 95% bound > 0 , at each of ≥ 2 capture budgets.

RQ3 (sim $\rightarrow$ real: collect few real rollouts). Which few real rollouts close the remaining gap, and how should they correct the twin and the policy?

H3. Real rollouts selected actively (by predicted gap or uncertainty) and used for twin and policy correction reach the target SR with $\geq 50\%$ fewer rollouts than random selection. *Decision:* $N_{\text{active}}(\tau_3) \leq 0.5 \cdot N_{\text{random}}(\tau_3)$ on fitted rollout-SR curves, upper 95% bound of the ratio < 1 ; τ_3 = SR of random selection at its largest budget.

RQ4 (whole loop: budget and generalization). What real-data budget does the full loop need compared with real-only learning, and does it hold beyond navigation?

H4. (a) The full loop (H1–H3 combined) reaches the target SR with $\leq 10\%$ of the real data needed by real-only learning (trained directly in the reference, every interaction counted as real data). *Decision:* $B_{\text{loop}}(\tau_4) \leq 0.1 \cdot B_{\text{real}}(\tau_4)$, upper 95% bound of the ratio < 0.2 ; τ_4 = SR of real-only learning at its largest budget. The budget curve ("exchange rate" of twin to real data) is also reported for a uniform loop and for a learned world model as the simulator. (b) With the pipeline and hyperparameters unchanged, on manipulation $B_{\text{loop}} / B_{\text{real}}$ has an upper 95% bound < 1 and the H1–H3 effects keep their sign.

Decision rule. A hypothesis holds when all its parts pass on tier A; a failed one is reported as a measured budget curve and does not block the next step, which has its own baseline.



8. Wkład spodziewanych wyników dla rozwoju dyscypliny naukowej / Contribution of the expected results to development of scientific discipline

Krótki opis spodziewanego wkładu wyników rozprawy doktorskiej dla rozwoju dyscypliny naukowej, w której realizowana jest rozprawa doktorska (maks. 1 strona, czcionka 11 pt., interlinia 1). / Brief description of the expected contribution of the results of the doctoral dissertation to the development of the scientific discipline in which the doctoral dissertation is carried out (at most 1 page, font 11, line spacing 1).

The dissertation is carried out in **information and communication technology** (*informatyka techniczna i telekomunikacja*), in the area of machine learning. Its object is the **real-to-sim-to-real learning loop** of embodied policies with digital twins built by neural reconstruction of real scenes, and how to make this loop work with **less real data**. Visual navigation is the primary testbed and robotic manipulation the cross-task test. Building twins, simulators or benchmarks, control design and mechanical engineering are not the object of the research; the student uses existing tools and public data.

Key original contribution. To our knowledge, the first treatment of the real data in the whole real-to-sim-to-real loop as **one budget to be allocated actively**, with a method for each step of the loop and a measured budget law for navigation, where no real-data-budget curves have been published.

Contributions to machine learning methodology:

1. **Capture less (RQ1, H1).** A task-aware capture method that chooses which real views to record for a twin from the reconstruction's uncertainty in the regions that matter for the policy, instead of for image quality alone, and a test of how much capture it saves against uniform and reconstruction-only view selection.
2. **Train robustly on an imperfect twin (RQ2, H2).** An uncertainty-aware training method that turns per-region reconstruction uncertainty and the representation distance to a small set of real images into augmentation and sample weights, compared with uniform domain randomization at an equal capture budget.
3. **Collect few real rollouts (RQ3, H3).** An active selection method for real rollouts, driven by the predicted twin-to-real gap and its uncertainty, together with a procedure that uses the selected rollouts to correct both the twin and the policy, compared with random selection.
4. **A budget law for the loop (RQ4, H4).** Budget–performance curves of the full loop that give the "exchange rate" between twin and real data (how much real data the loop needs, relative to learning from real data only, to reach a given real success rate), compared with a uniform loop and a learned world model used as the simulator, and a test of whether the loop carries over unchanged from navigation to manipulation.

Contributions in engineering terms (ITiT):

5. **Evaluation protocol and software (supports all RQs).** A proxy-reality protocol on public scene datasets that counts real data exactly without a robot (a high-fidelity reference as "reality", a low-budget twin as the simulator), and open-source code for the three methods and the budget curves, built on existing twin pipelines and simulators. It tells a practitioner how much to record and how many real trials to plan for a new site.

Significance for the discipline. Real target-domain data is the main cost of deploying learned systems, and models are increasingly trained on reconstructed or synthetic data and deployed on real sensor data. Methods that decide where real data is worth spending, and a measured law of how much of it a twin can replace, are relevant to active learning, learning under distribution shift and uncertainty estimation in general, beyond robotics.

Dissemination. Results are planned for peer-reviewed conferences worth 200 points and assigned to ITiT on the ministerial list of 5.01.2024: NeurIPS and ICLR (machine learning) and CVPR (computer



vision), with ICML, ICCV and ECCV for resubmissions, as in §3, §11 and §12. P1 (RQ1–RQ2, task-aware capture and uncertainty-aware training) targets NeurIPS 2027; P2 (RQ3, active real-rollout selection and twin correction) ICLR or CVPR 2028; P3 (RQ4, budget law and manipulation) NeurIPS 2028, with ICLR 2029 as the fallback.

9. Planowane metody badawcze / Planned research methods

Krótki opis metod badawczych, które są będą użyte w czasie realizacji rozprawy doktorskiej dla (maks. 2 strony, czcionka 11 pt., interlinia 1). / Brief description of the research methods that will be used during the implementation of the doctoral dissertation (at most 2 pages, font 11, line spacing 1).

The research follows the real-to-sim-to-real loop: real capture → twin → policy training in the twin → a few real rollouts → correction of the twin and the policy → deployment. Four stages (I–IV) match §3 and the research questions of §7: capture less (RQ1), train robustly on an imperfect twin (RQ2), collect few real rollouts (RQ3), and measure the budget of the whole loop (RQ4). For each stage we give the method, data, metric and success criterion. Twins are built with existing open-source 3D Gaussian Splatting (3DGS)

[2] pipelines and simulators; the student develops neither a simulator nor a benchmark. Visual navigation

(point-goal and image-goal) is the primary task; manipulation is the cross-task test in Stage IV.

Tier A protocol: proxy reality (decides the hypotheses). Real robots make "real data" slow to collect and hard to count, so the hypotheses are decided on public real scans. For ≥ 10 indoor scenes with dense real captures and reference laser scans (e.g. ScanNet++ [19]), a **reference twin** built from the full capture and the scan plays "reality", rendered in a GPU simulator (Habitat [1] or a 3DGS renderer such as GaussGym [5], chosen in T3.1). The **training twin** is a 3DGS reconstruction built from a subset of the same capture. This makes every unit of real data countable: *capture* = views selected from the recorded pool (converted to capture minutes at the recording rate), *real rollouts* and *real demonstrations* = episodes in the reference twin. Both are summed in one operator-time cost at a rate fixed in the pre-registration. Some scenes are held out for all tuning. The proxy's own error is reported: renders of the reference twin are compared with held-out real images (PSNR, LPIPS, depth error), and tier C checks that tier-A conclusions hold on a robot. *Tier B:* public real-world datasets and published paired sim/real results (SIMPLER [34]). *Tier C (validation only):* ≥ 2 PWR rooms captured with a phone or RGB-D camera and a mobile robot (planned cooperation with the K29 "Denali" Autonomous Robots Laboratory, agreement in T3.2). *Compute:* WCSS Lem (NVIDIA H100) and PLGrid allocations (to be applied for). Policies are trained by reinforcement learning (PPO) and by imitation of a shortest-path planner in the twin; small policies on frozen visual encoders keep the compute academic.

Stage I – task-aware capture (RQ1, H1; sem. 3–4).

- *Method:* per-region reconstruction uncertainty of the current twin (Fisher information [16] or a Bayes' Rays-type [18] field, adapted to 3DGS) is weighted by **task relevance**, i.e. how often a pilot policy trained in the current twin visits or attends to the region on sampled start–goal paths. The next views are chosen greedily where task-weighted uncertainty is highest, in rounds, until the budget is spent.
- *Baselines:* uniform subsampling of the capture, and reconstruction-only next-best-view (FisherRF-type [16]) at the same budget.
- *Metric:* budget–success-rate (SR) curves on tier A, with ≥ 4 capture budgets; capture needed to reach the SR that uniform capture reaches at the largest budget. Image fidelity (PSNR, LPIPS) is reported but is not the target.
- *Success criterion (H1):* $\geq 40\%$ fewer views than uniform capture at equal SR, and fewer than reconstruction-only selection (upper 95% bound of the ratio < 1), as in §7.

Stage II – uncertainty-aware training on an imperfect twin (RQ2, H2; sem. 4).



- *Method*: (a) augmentation whose strength follows per-region reconstruction uncertainty (appearance and geometry perturbations where the twin is unsure, little where it is sure); (b) sample weighting of twin frames by their representation distance (frozen DINOv2 [25] features) to a few real images. The images come from the capture itself, so they cost no extra real data.
- *Baselines*: uniform domain randomization [20] with tuned ranges, no randomization, and feature alignment [23] as an extra comparator; all at the same capture budget.
- *Metric and criterion (H2)*: tier-A SR on held-out scenes, paired by scene; ≥ 10 pp higher SR than uniform domain randomization at an equal capture budget (lower 95% bound > 0), at each of ≥ 2 budgets.

Stage III – few real rollouts and correction (RQ3, H3; end of sem. 4 – sem. 5).

- *Selection*: candidate real rollouts (start–goal pairs) are scored by the predicted twin-to-real gap: disagreement of a small policy ensemble, plus the reconstruction uncertainty along the planned path. The top-k are executed in reality (tier A: the reference twin).
- *Use*: the rollouts (i) correct the twin, by re-capturing or re-weighting the regions where real and twin observations disagree (photometric correction in the style of [35]), and (ii) fine-tune the policy by co-training with the real rollouts [13]. Real SR is estimated from few trials with prediction-powered inference [32, 33].
- *Baselines*: random selection and uniform coverage of the same number of rollouts; related selection rules [27–29] as comparators where they apply.
- *Success criterion (H3)*: the target SR is reached with $\geq 50\%$ fewer real rollouts than random selection (upper 95% bound of the ratio < 1).

Stage IV – budget of the whole loop and cross-task test (RQ4, H4; sem. 6).

- *Budget curve*: total real data (capture + rollouts, one operator-time cost) against real SR for the full loop (Stages I–III) and for **real-only learning**, i.e. a policy trained only on data collected in the reference with the same encoder and algorithm, every interaction counted as real data. The "exchange rate" is the ratio of the budgets that reach the target SR; real-only scaling results [9, 10] give the expected curve shapes. The curve is also reported for a uniform loop (no Stage I–III guidance) and for a frozen, pretrained world model used as the simulator (comparator only; not trained).
- *Manipulation*: the same pipeline, unchanged, on public manipulation scenes with twin environments (ManiSkill3 [8]) under the tier-A protocol; published paired sim/real results (SIMPLER [34]) check that the proxy ranks policies as reality does (tier B).
- *Success criterion (H4)*: the loop reaches the target SR with $\leq 10\%$ of the real data needed by real-only learning (upper 95% bound of the ratio < 0.2); on manipulation, with pipeline and hyperparameters unchanged, the ratio's upper bound is < 1 and the H1–H3 effects keep their sign.

Tier C validation. Two robot campaigns (sem. 5 and 7) in 2 PWr rooms: phone or RGB-D capture, twins, policies with uniform capture versus the loop, and the few selected real rollouts. About 8 robot-hours per campaign (3 policies $\times$ 2 rooms $\times$ 30 episodes at ~ 2 min each, plus 2×20 selected rollouts). Tier C is reported as agreement with tier A and is never used to tune thresholds.

Statistics and reproducibility. Tests are one-sided ($\alpha = 0.05$), bootstrapped over scenes and seeds, Holm-corrected within each hypothesis, as in §7. The protocol is pre-registered in the project repository before each stage, and changes are logged with reasons. Code and scripts are released where the dataset licences permit.

10. Streszczenie popularnonaukowe / Abstract for general public

Streszczenie popularnonaukowe należy przygotować w dwóch językach: polskim i angielskim. Wersje językowe muszą być tożsame. Opis powinien być napisany językiem popularnonaukowym i zawierać cel rozprawy, opis planowanych badań naukowych oraz powodów, dla których podjęta została ta tematyka badawcza, a także najważniejsze spodziewane efekty (maks. 1 strona, czcionka 11 pt., interlinia 1). / *The description for general public must be in Polish and in English. The language*



versions must be identical. The description should be written for general public and should include the doctoral dissertation goal, description of research, reasons for attempting a particular research topic and substantial results expected (at most 1 page, font 11, line spacing 1).

Streszczenie popularnonaukowe

Roboty uczą się dziś samodzielnego działania, na przykład poruszania się po budynku lub chwytania przedmiotów, metodą prób i błędów. Potrzebują do tego bardzo wielu prób, więc zwykle ćwiczą w symulacji komputerowej. Coraz częściej jest nią cyfrowy bliźniak: wierna, fotorealistyczna kopia prawdziwego miejsca, którą metody sztucznej inteligencji potrafią odtworzyć z krótkiego nagrania. Powstaje w ten sposób pętla: nagrywamy prawdziwe miejsce, budujemy jego kopię, uczymy w niej robota, sprawdzamy go w rzeczywistości, poprawiamy i dopiero wtedy wdrażamy. Każdy z tych kroków wymaga jednak prawdziwych danych, czyli nagrań i prób w prawdziwym świecie, a ich zbieranie jest powolne i kosztowne.

Celem rozprawy jest sprawdzenie, jak przejść przez tę pętlę z jak najmniejszą ilością prawdziwych danych. Zamiast nagrywać wszystko po równo, będziemy nagrywać przede wszystkim te miejsca, które są ważne dla zadania robota i których kopia jest jeszcze niepewna. Ponieważ kopia nigdy nie jest idealna, nauczymy robota tak, aby nie polegał na jej niepewnych fragmentach i radził sobie z jej błędami. Na koniec wybierzemy tylko te nieliczne próby w rzeczywistości, które najwięcej mówią o tym, gdzie robot jeszcze zawodzi, i wykorzystamy je do poprawienia zarówno kopii, jak i robota. Wskazówek dostarczy sama kopia, która wie, gdzie jest niepewna, oraz sposób, w jaki model sterujący robotem opisuje to, co widzi. Głównym zadaniem testowym będzie nawigacja robota mobilnego, a sprawdzianem uniwersalności chwytanie przedmiotów ramieniem robota. Badania będą prowadzone przede wszystkim na publicznie dostępnych nagraniach i trójwymiarowych skanach prawdziwych budynków, a wybrane wyniki zostaną potwierdzone na prawdziwym robocie.

Spodziewanym efektem są metody, które pozwolą nagrywać mniej, uczyć skuteczniej i wykonywać mniej kosztownych prób w rzeczywistości, oraz pomiar, ile prawdziwych danych potrzeba, gdy robot uczy się w cyfrowej kopii. Może to przyspieszyć i potanić wdrażanie robotów w magazynach, szpitalach, biurach i fabrykach. Opracowane oprogramowanie zostanie udostępnione publicznie.

Abstract for general public

Robots today learn to act on their own, for example to move around a building or to grasp objects, by trial and error. They need a very large number of trials, so they usually practise in a computer simulation. Increasingly, this is a digital twin: a faithful, photorealistic copy of a real place that artificial intelligence methods can reconstruct from a short recording. This creates a loop: we record the real place, build its copy, train the robot in it, check it in reality, correct it and only then deploy it. Each of these steps, however, needs real data, that is, recordings and trials in the real world, and collecting them is slow and costly.

The goal of this dissertation is to find out how to go through this loop with as little real data as possible. Instead of recording everything evenly, we will record mainly the places that matter for the robot's task and whose copy is still uncertain. Because the copy is never perfect, we will train the robot so that it does not rely on the uncertain parts of the copy and copes with its errors. Finally, we will choose only the few trials in reality that tell the most about where the robot still fails, and use them to correct both the copy and the robot. Guidance will come from the copy itself, which knows where it is uncertain, and from the way the model that controls the robot describes what it sees. Mobile-robot navigation will be the main test task, and grasping objects with a robot arm will test how universal the approach is. The research will be carried out mainly on publicly available recordings and three-dimensional scans of real buildings, and selected results will be confirmed on a real robot.

The expected result is methods that make it possible to record less, train more effectively and carry out fewer costly trials in reality, and a measurement of how much real data is needed when a robot



learns in a digital copy. This may make deploying robots in warehouses, hospitals, offices and factories faster and cheaper. The software developed will be made publicly available.

11. Termin oddania do druku artykułu naukowego lub monografii / Date of submission for publication of at least one scientific article or monography

Termin wysłania do recenzji artykułu naukowego lub monografii naukowej zgodnie z wymaganiami art. 186 ust. 1 pkt 3 Ustawy stawianymi osobie ubiegającej się stopień doktora. Termin dotyczy oddania do druku co najmniej jednego artykułu naukowego opublikowanego w czasopiśmie naukowym lub w recenzowanych materiałach z konferencji międzynarodowej w odpowiedniej dyscyplinie naukowej, które w roku opublikowania artykułu w ostatecznej formie były ujęte w wykazie („lista ministerialna”) sporządzonym zgodnie z przepisami wydanymi na podstawie Ustawy (art. 267 ust. 2 pkt 2 lit. b) lub 1 monografii naukowej wydanej przez wydawnictwo, które w roku opublikowania monografii w ostatecznej formie było ujęte w wykazie sporządzonym zgodnie z przepisami wydanymi na podstawie Ustawy (art. 267 ust. 2 pkt 2 lit. a). / A date of submission for publication of at least one scientific article published in a scientific journal or in the peer-reviewed materials of an international conference in the relevant scientific discipline, which in the year of publication of the article in its final form were included in the list drawn up in accordance with the regulations issued under the Act (Art. 267 sec. 2 point 2(b) or 1 scientific monograph published by a publishing house, which in the year of publication of the monograph in its final form was included in the list drawn up in accordance with the regulations issued pursuant to the Act (Art. 267 sec. 2 point 2(a), referred to in Art. 186 sec. 1 point 3 of the Act.

May 2027 / maj 2027

Paper **P1** submitted for review to the main track of the **Conference on Neural Information Processing Systems (NeurIPS 2027)**. NeurIPS is peer-reviewed international conference proceedings worth **200 points** on the ministerial list (conference Lp. 87), assigned to the discipline *informatyka techniczna i telekomunikacja*. P1 reports Stages I–II (RQ1–RQ2): task-aware, uncertainty-guided capture of digital twins and uncertainty-aware training of policies on imperfect twins, evaluated by the real data they save (task T4.2). If it is rejected, the revised paper goes to **CVPR 2028** (expected deadline November 2027, also 200 points), and then to the next 200-point conference (§12).

12. Inne / Other comments

Dodatkowe wyjaśnienia, komentarze, podsumowanie związane z indywidualnym planem badawczym (maks. 1 strona, czcionka 11 pt., interlinia 1). / Additional explanations, comments, summary related to the individual research plan (at most 1 page, font 11, line spacing 1).

Planned outputs by year (200-point conferences assigned to the discipline, list of 5.01.2024).

Year	Papers (planned submission, 200 points each)	Code	Grants	Mobility, events
1 (2025/26)	—	—	—	—
2 (2026/27)	P1 (RQ1–RQ2, task-aware capture, uncertainty-aware training): NeurIPS 2027, May 2027 (§11)	Loop on tier A	SzD Minigrant; NCN PRELUDIUM; NAWA Bekker	Summer school poster



3 (2027/28)	P2 (RQ3, few real rollouts, twin correction): CVPR or ICLR 2028; P3 (RQ4, budget of the full loop, manipulation): NeurIPS 2028	Rollout selection	—	Mid-term; 3-month visit (sem. 6)
4 (2028/29)	Optional P4, consolidated study: ICLR 2029 or CVPR 2029	Full release	—	Dissertation

P1–P3 are the linked core of the dissertation; P4 is optional.

Foreign research visit (candidate hosts, not yet contacted). (1) Z. Kira's group, Georgia Tech: EmbodiedSplat [3]. (2) Multi-robot Systems Lab, Stanford University (M. Schwager): Gaussian-splatting scene models and navigation. A European host (Erasmus+ route) working on active reconstruction or sim-to-real transfer, and a foreign co-author for P2, are sought with the supervisor in sem. 4.

Collaboration at PWr. Robot validation is planned with the K29 *Denali* laboratory (Pioneer 3-DX, Jaguar 4x4, ROS 2); a written agreement is task T3.2 (Nov 2026).

Risks and mitigation (semester affected).

- *Task-aware capture gives no gain* (3–4): the measured capture–SR curves still answer RQ1, and Stages II–III have their own baselines (§7).
- *3DGS uncertainty is poorly calibrated* (3–4): two estimators and an ensemble fallback, checked against the reference twin.
- *Proxy reality is easier than reality* (4–7): the proxy's own error is reported; tier C checks the direction of the effects.
- *Real-only learning is strong at small budgets* (6): gains saturate quickly in a known location [10]; H4 is reported as a full budget curve.
- *Scooping* (3–6): twin-based robot learning moves fast [3–7, 11]; early preprints, monthly monitoring.
- *Dataset licence or robot access* (3, 5): other public scans with reference geometry, own phone captures of PWr rooms; without a robot, tier C moves by a semester and hypotheses are still decided on tier A.
- *Rejection at a top venue* (4–7): each paper has a resubmission path inside the 200-point set 2–4 months later (NeurIPS → CVPR → ICML → ECCV/ICCV).

Prior work (background only). A pre-PhD NLP paper (ACL 2025 SRW, forecasting from hidden states); not part of the dissertation.

Ethics and data. Own captures are anonymized (GDPR); public datasets are used under their licences.

13. Określenie planowanej formy współpracy z promotorem / Planned form of cooperation with the supervisor

Krótki opis planowanej formy współpracy z promotorem, np. częstotliwość spotkań, forma spotkań i ich zakres, sposób komunikacji. / Brief description of the planned form of cooperation with the supervisor, e.g., frequency of meetings, form of meetings and their scope, method of communication.

- **Weekly meeting** (~1 h, in person at K46 or online via MS Teams) to discuss progress, results and next steps.
- **Monthly written progress summary** (completed tasks, results, problems, plan). It lists every IPB task from §3 with its % **completion**, in the same form as the task table of the mid-term self-report.



- **Semester review** against the IPB schedule (§3), before each annual report and the mid-term evaluation. Deviations from the plan and their reasons are recorded and agreed with the supervisor.
- **Supervisor co-signs** the annual reports and the mid-term self-report (autoreferat).
- **Paper-writing sprints** before each submission deadline, with joint editing in Overleaf. The supervisor co-authors and reviews the papers before submission.
- **Ongoing communication** via e-mail / Teams chat. The code and experiment logs are in a shared Git repository that the supervisor can access.

14. Opinia promotora pomocniczego / Opinion of assistant supervisor

Promotor pomocniczy (jeżeli jest wyznaczony) powinien zaopiniować IPB. / Assistant supervisor (if appointed) should provide an opinion of the IRP.

nie dotyczy / not applicable

.....
Podpis promotora pomocniczego /
Signature of assistant supervisor

15. Podpisy doktoranta i promotora/promotorów / Signatures of doctoral student and supervisor(s)

Wybierz datę/Select date

.....
Data / Date

.....
Podpis doktoranta / Signature of doctoral student

.....
Podpis promotora / Signature of supervisor

.....
Podpis drugiego promotora / Signature of second supervisor